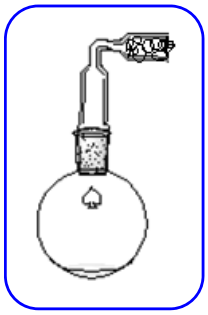
Inorganic experiment 1: Homogenous Catalysis

Session III: Study of Catalytic Reaction: $RuCl_2(PPh_3)_3$-catalysed Oxidation of 1-phenylethanol to acetophenone

1. Add 8 mL of dried acetone and a magnetic stirrer bar into a 100 mL round bottom flask.
2. Using a micropipette, add 0.06 mL of 1-phenylethanol into the reaction flask upon stirring.
3. Upon stirring, add 0.07 g of K_2CO_3 into the reaction flask.
4. Lastly, add 0.06 g of your $RuCl_2(PPh_3)_3$ obtained into the reaction flask upon stirring, and then quickly connect a water-cooled reflux condenser with a calcium chloride drying tube to the reaction flask containing the reactants.
5. Reflux the reaction mixture for **60 minutes**. *[Note: The temperature of oil bath is < 100 °C.]*
6. Cool the reaction mixture to room temperature after the reflux.
7. Upon stirring, add 50 mL of *n*-hexane into the reaction flask to quench the reaction.



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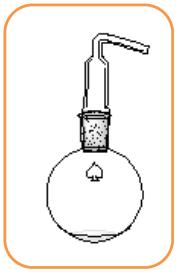
Inorganic experiment 1: Homogenous Catalysis

Session III: Study of Catalytic Reaction: $RuCl_2(PPh_3)_3$-catalysed Oxidation of 1-phenylethanol to acetophenone

8. Using a Hirsch funnel, filter the reaction mixture through Celite by suction filtration and collect the filtrate. *[Note: The suction flask must be washed and dried completely before use. **When the suction is on, a thin layer of Celite should be packed on top of the filter paper in the Hirsch funnel.**]*
9. Transfer the filtrate from the suction flask into a 150 mL round bottom flask. Rinse the suction flask with *n*-hexane (2 x 5 mL) and transfer it into the round bottom flask.
10. Using a rotary evaporator, remove the solvent from the reaction mixture in *n*-hexane under reduced pressure.
11. Connect the round bottom flask with a gas adapter and dry the oily product under vacuum for 15 - 20 minutes.
12. Using a micropipette, add 0.07 mL of mesitylene into the reaction flask upon stirring.
13. Add 0.6 mL of $CDCl_3$ into the round bottom flask containing oily product and transfer all solution into an NMR tube with a clean glass dropper. Label the NMR tube clearly and submit it for ^1H NMR analysis.



Celite



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Inorganic experiment 1: Homogenous Catalysis

Session III: Study of Catalytic Reaction: $RuCl_2(PPh_3)_3$-catalysed Oxidation of 1-phenylethanol to acetophenone

At the end of this session:

(i) Submit the rest of your product, $RuCl_2(PPh_3)_3$ [obtained in session I] if it is available;

(ii) Submit the oily product(s) [obtained after session III of experiment 1] for ^1H NMR analysis **in $CDCl_3$ in a NMR tube**;

(iii) Record the raw data and observation on your own datasheet (signed by your GTA).

Reminder: Take note on the proforma requirement [please refer to the Inorganic notes p.22].

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Inorganic experiment 1 (session III): Risk assessment

Experiment-Based Risk Assessment Form

Name of Department	Department of Chemistry	Location of Lab	S5-04
Name of Laboratory	Chemical Synthesis Teaching Lab	Name of PI (lecturer-in-charge)	Dr. Edith S. H. Chan
Name of LO	CM3191 STUDENTS	Name of Activity/Experiment	CM3191 Inorganic - Homogenous Catalysis: Session III - Study of Catalytic Reaction: RuCl2(PPh3)3-catalysed Oxidation of 1-phenylethanol to

Hazard Identification				Risk Evaluation & Control						
No	Description / Details of Steps in Activity	Hazard(s)	Possible Accident(s)	Existing Risk Control (Mitigation)	Severity	Likelihood (probability)	Risk Level	Additional Risk Control	Person Responsible	By (Date)
1	Clean and dry all the glassware (round-bottomed flask, conical flasks, beakers, etc) used in the reactions	Broken Glassware	Cuts from broken glassware	Visual inspection for any cracks before usage. Proper disposal of broken glassware in Sharps container.	2	1	2			
2	Add 8 mL of acetone and a magnetic stirrer bar into a 100 mL round bottom flask. Using micropipette, add 0.06 mL of 1-phenylethanol into the reaction flask upon stirring.	**Chemical hazard:** 1-phenylethanol is harmful and an irritant.	Harmful if swallowed. Cause eye irritation and potential severe eye damage	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use	2	1	2			
		Chemical and fire hazards: Acetone is harmful, flammable and irritating	Harmful if swallowed. Cause skin and respiratory tract irritation. Acetone is flammable.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use 4. Keep acetone away from open or naked flame and a distance away from the hotplate 5. In case of fire, use dry sand, dry chemical or alcohol resistant foam or water spray or carbon dioxide for extinction.	2	1	2			
3	Upon stirring, add 0.07 g of K2CO3 into the reaction flask containing 1-phenylethanol and acetone. Lastly, add 0.06 g of RuCl2(PPh3)3 obtained into the reaction flask upon stirring, and then quickly connect a water-cooled reflux condenser with a calcium chloride (CaCl2) drying tube to the reaction flask containing the reactants.	**Chemical hazards:** RuCl2(PPh3)3, 1-phenylethanol, K2CO3 and CaCl2 are harmful and irritating	Harmful if swallowed. Cause skin, eye and respiratory tract irritation	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use	2	1	2			
		Chemical and fire hazards: Acetone is harmful, flammable and irritating	Harmful if swallowed. Cause skin and respiratory tract irritation. Acetone is flammable.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use 4. Keep acetone away from open or naked flame and a distance away from the hotplate 5. In case of fire, use dry sand, dry chemical or alcohol resistant foam or water spray or carbon dioxide for extinction.	2	1	2			

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4	Reflux the reaction mixture (containing 1-phenylethanol, mesitylene RuCl2(PPh3)3, K2CO3 and acetone) for 60-120 minutes. Cool the reaction mixture to room temperature after the reflux is completed. Upon stirring, add 50 mL of n-hexane into the reaction flask to quench the reaction.	**Physical and heat hazards**: Usage of hot oil bath and hotplate during reflux	Spillage of oil bath. Hot oil in oil bath and hot surface of hot plate can cause scalding of skin.	1. Ensure appartus is properly set-up and clamped and condenser is not leaking water into the oil bath. 2. Turn off the hotplate after reflux is completed	2	1	2			
		Chemical and fire hazards: Acetone and *n*-hexane are harmful, flammable and an irritant	Harmful if swallowed. Cause eye irritation. Acetone and *n*-hexane are flammable. Spillage of acetone and *n*-hexane during the transfer process.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use 4. Keep acetone away from open or naked flame and a distance away from the hotplate 5. In case of fire, use dry sand, dry chemical or alcohol resistant foam or water spray or carbon dioxide for extinction.	2	1	2			
		Chemical hazard: 1-phenylethanol is harmful and an irritant.	Harmful if swallowed. Cause eye irritation and potential severe eye damage	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use	2	1	2			
		Chemical hazard: RuCl2(PPh3)3 is harmful and an irritant	Harmful if swallowed. Cause skin and respiratory tract irritation	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use	2	1	2			
		Chemical hazards: K2CO3 and CaCl2 are harmful and an irritant	Harmful if swallowed. Cause skin, eye and respiratory tract irritation	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use	2	1	2			

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5	Using a Hirsch funnel, filter the reaction mixture (containing n-hexane, acetophenone, 1-phenylethanol and RuCl2(PPh3)3) through Celite by suction filtration and collect the filtrate. Transfer the filtrate from the suction flask into a 150 mL round bottom flask. Rinse the suction flask with n-hexane (2 x 5 mL) and transfer it into the round bottom flask.	**Chemical hazard**: Celite is harmful and can cause skin and eye irritation	Harmful if swallowed May cause irritation to the skin, eye or respiratory systems	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use	2	1	2			
		Chemical and fire hazards: *n*-hexane is harmful, flammable and can cause skin irritation	Harmful if swallowed. Cause skin irritation Hexane is flammable	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use 4. Keep hexane away from open or naked flame and a distance away from the hotplate 5. In case of fire, use dry sand, dry chemical or alcohol resistant foam or water spray or carbon dioxide for extinction.	2	1	2			
		Chemical hazard: Acetophenone is harmful and can cause eye irritation or damage	Harmful if swallowed. Cause eye irritation and potential severe eye damage	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use	2	1	2			
		Chemical hazard: 1-phenylethanol is harmful and an irritant.	Harmful if swallowed. Cause eye irritation and potential severe eye damage	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use	2	1	2			
		Chemical hazard: RuCl2(PPh3)3 is harmful and an irritant	Harmful if swallowed. Cause skin and respiratory tract irritation	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use	2	1	2			
		Chemical hazards: K2CO3 and CaCl2 are harmful and an irritant	Harmful if swallowed. Cause skin, eye and respiratory tract irritation	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use	2	1	2			
		Mechanical & Physical hazards: Usage of set-up for suction filtration	Glassware might topple down when the vacuum is used and could break, resulting in cuts.	1. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 2. Visually inspect apparatus for cracks before use 3. Clamp the apparatus properly before usage	2	1	2			

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6	Using a rotary evaporator, remove the solvent from the reaction mixture in n-hexane under reduced pressure. Connect the round bottom flask with a gas adapter and dry the oily product under vacuum for 15 - 20 minutes.	**Chemical hazard:** Acetophenone is harmful and can cause eye irritation or damage	Harmful if swallowed. Cause eye irritation and potential severe eye damage	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use	2	1	2			
		Chemical hazard: 1-phenylethanol is harmful and an irritant.	Harmful if swallowed. Cause eye irritation and potential severe eye damage	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use	2	1	2			
		Chemical and fire hazard: n-hexane is harmful, flammable and can cause skin irritation	Harmful if swallowed. Cause skin irritation Hexane is flammable	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use 4. Keep hexane away from open or naked flame and a distance away from the hotplate	2	1	2			
		Electrical hazard: Use of rotary evaporator	Potential electric shock from the use of rotary evaporator	1. Ensure the rotary evaporator is properly grounded and area around it is not wet 2. In case of fire, use dry sand, dry chemical or alcohol resistant foam or water spray or carbon dioxide for extinction.	2	1	2			
		Physical hazard: Broken glasswares	Injuries from broken glasswares	Glasswares inspection before usage and handle glasswares with care.	2	1	2			
7	Using micropipette, add 0.07 mL of mesitylene into the reaction flask upon stirring.	**Chemical hazard:** 1-phenylethanol is harmful and an irritant.	Harmful if swallowed. Cause eye irritation and potential severe eye damage	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use	2	1	2			
		Chemical hazard: Acetophenone is harmful and can cause eye irritation or damage	Harmful if swallowed. Cause eye irritation and potential severe eye damage	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use	2	1	2			
		Chemical and fire hazards: Mesitylene is harmful, flammable and irritating	Harmful if swallowed. Cause skin and respiratory tract irritation. Mesitylene is flammable.	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use 4. Keep mesitylene away from open or naked flame and a distance away from the hotplate 5. In case of fire, use dry sand, dry chemical or alcohol resistant foam or water spray or carbon dioxide for extinction.	2	1	2			

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8	Add 0.6 mL of CDCl3 into the round bottom flask containing oily product and transfer all solution into a NMR tube with a clean glass dropper.	**Physical hazard**: NMR tube and glass dropper	Injury from broken NMR tube or glass dropper	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use 4. Handle the NMR tube and glass dropper with care and do not use too much force when caping the NMR	2	1	2			
		Chemical Hazard: Deuterated chloroform (CDCl3)	Deuterated chloroform is carcinogenic and an irritant to the skin and eyes	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment.	2	1	2			
		Chemical hazard: Acetophenone is harmful and can cause eye irritation or damage	Harmful if swallowed. Cause eye irritation and potential severe eye damage	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use	2	1	2			
		Chemical hazard: 1-phenylethanol is harmful and an irritant.	Harmful if swallowed. Cause eye irritation and potential severe eye damage	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use	2	1	2			
		Chemical hazard: RuCl2(PPh3)3 is harmful and an irritant	Harmful if swallowed. Cause skin and respiratory tract irritation	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use	2	1	2			
		Chemical and fire hazards: Mesitylene is harmful, flammable and an irritant	Harmful if swallowed. Cause skin and respiratory tract irritation Mesitylene is flammable	1. Conduct the experiment in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Visually inspect apparatus for cracks before use 4. Keep mesitylene away from open or naked flame and a distance away from the hotplate 5. In case of fire, use dry sand, dry chemical or alcohol resistant foam or water spray or carbon dioxide for extinction.	2	1	2			
9	Washing of glasswares with acetone after experiment is completed	**Chemical and fire hazard:** Acetone is an irritant and is flammable	Can cause irritation to skin and eyes. Flammable substance that can start a fire.	1.Wash the apparatus in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Avoid placing ochenne near any open or naked flame 4. Dispose the washing solvent (acetone) and chemical used in each apparus into the appropriate chlorinated, non-chlorinated or heavy metal waste solvent bottle 5. In case of fire, use dry sand, dry chemical or alcohol resistant foam or water spray or carbon dioxide for extinction.	2	1	2			
		Physical hazard: Broken glasswares	Cuts and injury from broken glasswares	1.Wash the apparatus in a well ventilated fumehood 2. Wear suitable PPE (lab coat, googles, gloves and covered shoes) when conducting experiment. 3. Handle the glasswares with care to avoid dropping them 4. Dispose the broken glasswares into the appropriate Sharps bin	2	1	2			

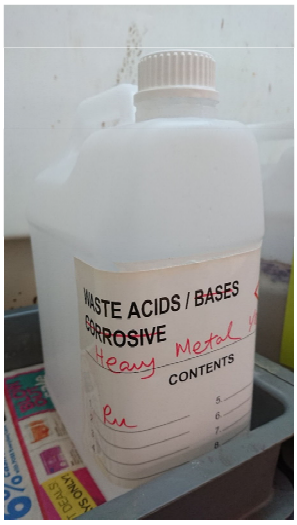
Conducted By	**Approved By**
Name	Name Dr. Edith Chan
Signature	Signature
Date	Approval date Next Revision date

Dried solvent



< REMINDER >

Ru (heavy metal) waste bottle



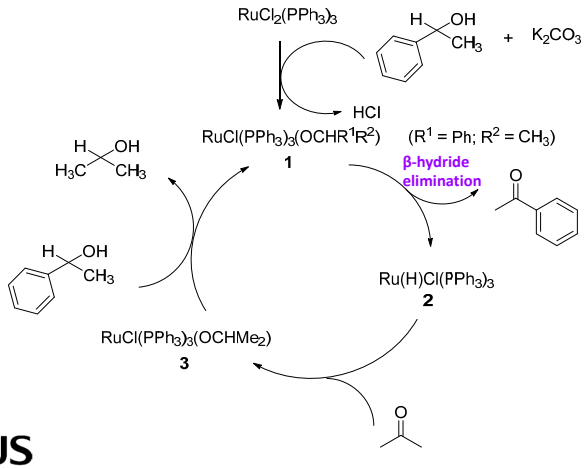


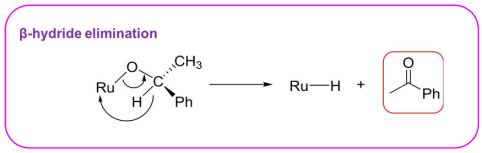
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Inorganic experiment 1: Homogenous Catalysis

Session III: Study of Catalytic Reaction: $RuCl_2(PPh_3)_3$-catalysed Oxidation of 1-phenylethanol to acetophenone

- Proposed mechanism for the $RuCl_2(PPh_3)_3$-catalyzed oxidation of 1-phenylethanol to acetophenone
  (source: *J. Chem. Educ.* **2011**, *88*, 657–661)





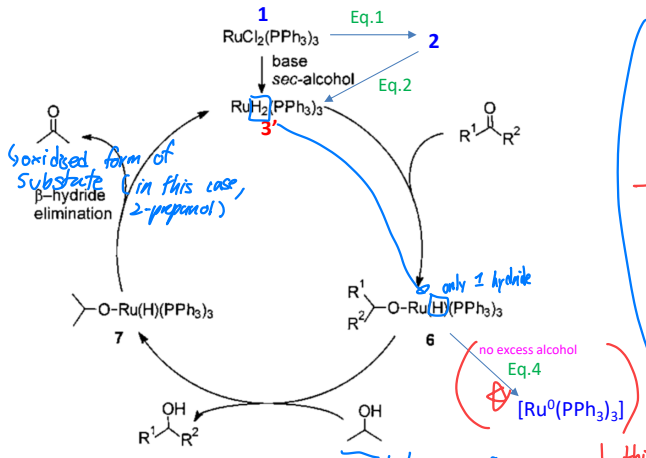
References:
(i) *J. Chem. Educ.* **2011**, *88*, 657–661
(ii) *J.-E. Chem.- Eur. J.* **1996**, *2*, 1533–1536

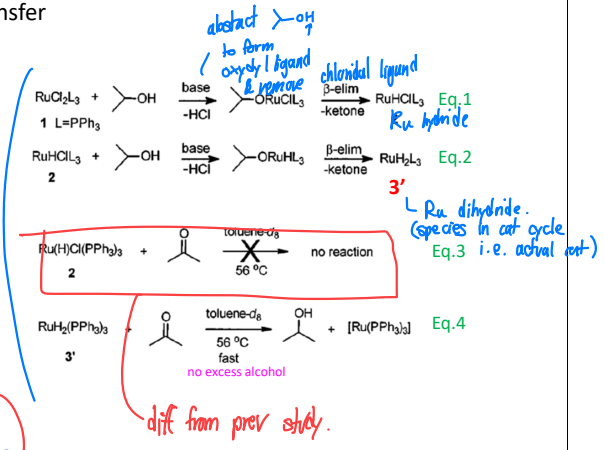
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Inorganic experiment 1: Homogenous Catalysis

Session III: Study of Catalytic Reaction: $RuCl_2(PPh_3)_3$-catalysed Oxidation of 1-phenylethanol to acetophenone

- Proposed mechanism for the $RuCl_2(PPh_3)_3$-catalyzed hydrogen transfer
  (source: *Chem. Commun.*, **1999**, *351–352*)

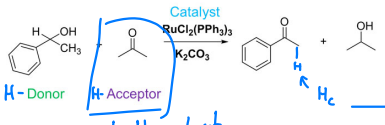




Reference: *Chem. Commun.*, **1999**, 351–352

* why no need to measure empty rbf mass?
↳ many intermediates. ✗ know at which step rxn is quenched
↳ since no purification, resulting mass ≠ intended product mass
↳ use internal standard + NMR to quantify intended product

Acetophenone



└ the solvent for exp 1-2 is actually act as acceptor ☆

Spectral Database for Organic Compounds SDBS

SDBS Information

SDBS No.: 722

Compound Name: acetophenone

Molecular Formula: C_8H_8O

Molecular Weight: 120.1

CAS Registry No.: 98-86-2

Spectral Code:
Mass ;
^{13}C NMR : in $CDCl_3$
^{13}C NMR : in $CDCl_3$
^1H NMR : 90 MHz in $CDCl_3$
^1H NMR : in $CDCl_3$
IR : liquid film
IR : CCl_4 solution
ESR : ANION RADICAL, ELECTROLYTIC REDUCTION IN DMF
ESR : ANION RADICAL, ELECTROLYTIC REDUCTION IN DME

Chemical Information;

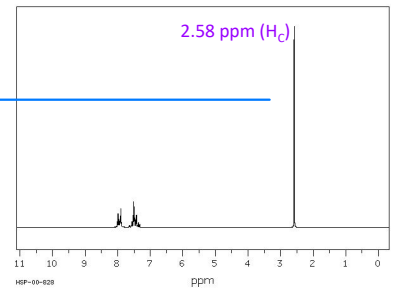
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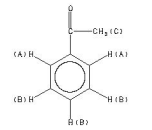
URL for this Compound:
https://sdbs.db.aist.go.jp/sdbs/cgi-bin/landingpage?sdbsno=722

External Information:
external link displays in a separate page

- Japan Chemical Information Link Center (in Japanese)
  - JST Nikkaji Web (in Japanese)

SDBS-^{1}H NMRSDBS No. 722HSP-00-828
C_8H_8O
acetophenone





Assign.	Shift(ppm)
A	7.94
B	7.68 to 7.32
C	2.585

peak data

SDBS No. 722HSP-00-828

Source of spectrum: https://sdbs.db.aist.go.jp/sdbs/cgi-bin/direct_frame_top.cgi

Mesitylene

- as internal standard ☆

└ can be used as it's inactive (x react ⇒ product)

Spectral Database for Organic Compounds SDBS

SDBS Information

SDBS No.: 1351

Compound Name: mesitylene

Molecular Formula: C_9H_{12}

Molecular Weight: 120.2

CAS Registry No.: 108-67-8

Spectral Code:
Mass ;
^{13}C NMR : in $CDCl_3$
^1H NMR : 90 MHz in $CDCl_3$
IR : liquid film
IR : CCl_4 solution
Raman : 4880 A,200 M,liquid

Chemical Information;

Return to Search;
Return to Result;

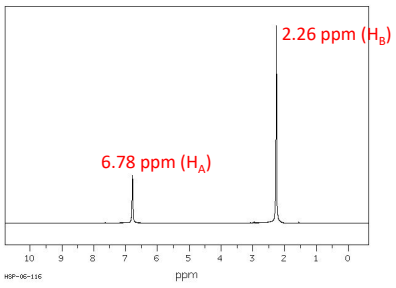
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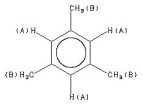
External Information:
external link displays in a separate page

- Japan Chemical Information Link Center (in Japanese)
  - JST Nikkaji Web (in Japanese)

SDBS-^{1}H NMRSDBS No. 1351HSP-06-116
C_9H_{12}
mesitylene

89.56 MHz





Assign.	Shift(ppm)
A	6.78
B	2.26

peak data

SDBS No. 1351HSP-06-116

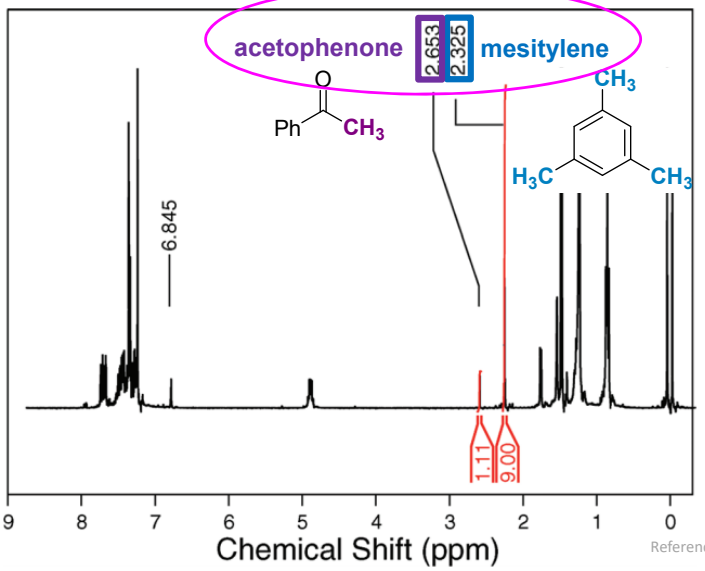
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^{1}H-NMR Characterisation

Mesitylene and Acetophenone

WHY ???

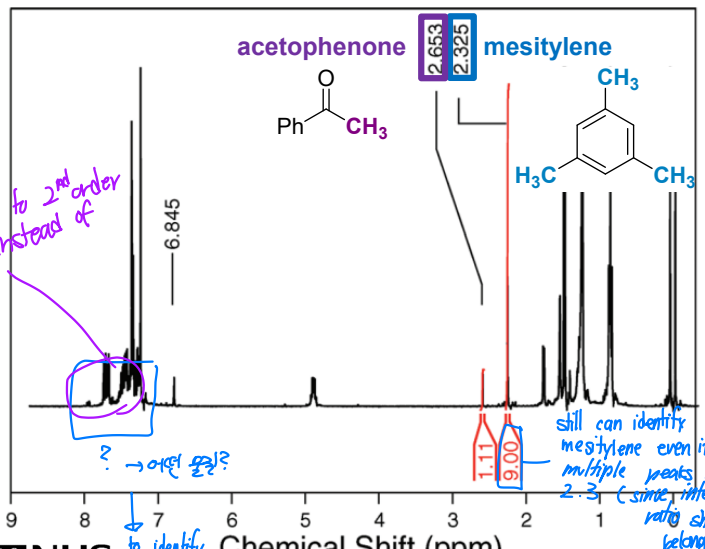
for report, integrate All peaks
→ give hints on rxn (maybe which stage rxn is quenched?)



Reference: *J. Chem. Edu.* 2011, 88, 657-661.

^{1}H-NMR Characterisation

Mesitylene and Acetophenone



maybe due to 2nd order spectrum instead of impurity

? → 아닌 것같은?

still can identify mesitylene even if multiple peaks around 2.3 (since integration ratio shows # of H belonging to the peak)

to identify, note filtrate colour & mesitylene colour (compare exp 1-2 & 1-3 to get clearer idea)

$$\frac{n_A}{n_B} = \frac{\left(\frac{integral\ of\ peak\ X\ from\ A}{no.\ of\ contributing\ nuclei\ to\ peak\ X}\right)}{\left(\frac{integral\ of\ peak\ Y\ from\ B}{no.\ of\ contributing\ nuclei\ to\ peak\ Y}\right)}$$

$$\frac{n_{acetophenone}}{n_{mesitylene}} = \frac{\left(\frac{1.11}{3}\right)}{\left(\frac{9}{9}\right)} = 0.37$$

Amount known

If *z* moles of mesitylene is used,

$n_{acetophenone}$ = 0.37*z* mol

Mass of acetophenone = 0.37*z* × 120.15
= 44.5*z* g

Reference: *J. Chem. Edu.* 2011, 88, 657-661.

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Turnover Frequency (TOF)

- To determine the activity of a catalyst
- More active catalyst → higher TOF

only applicable to exp 1-3
as 1-2 don't use cat

$$TOF = \frac{\left[\dfrac{\text{amount of product (acetophenone)}}{\text{amount of catalyst used } (RuCl_2(PPh_3)_3)}\right]}{\text{time}}$$

in mols

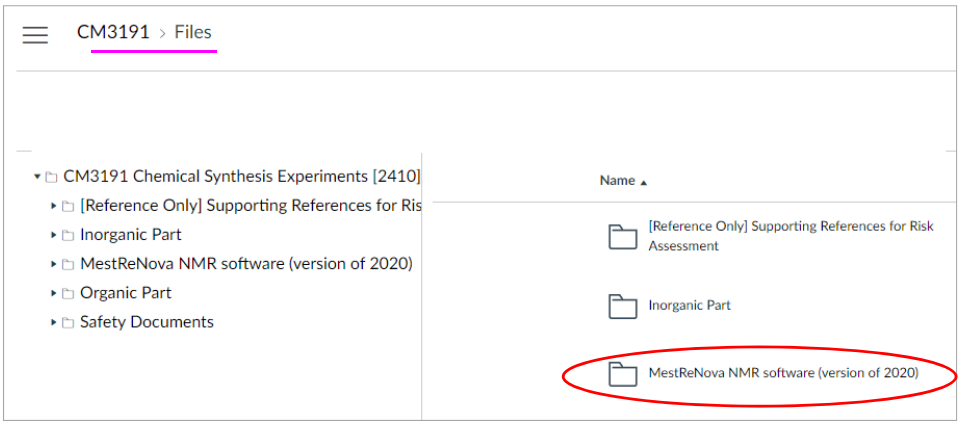
in mins

Remarks: Your result of TOF and its discussion are required in your lab report.

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NMR software: MestReNova

- GTA for NMR: Nie Ziwei



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Dinuclear Ru catalyst in oxidation of alcohols

CHEMISTRY
A European Journal
ChemPubSoc Europe

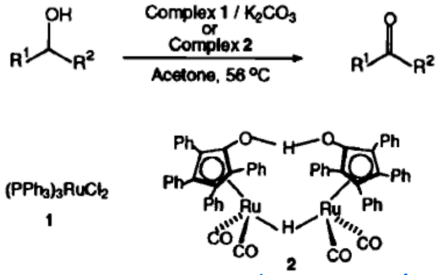
Volume 2, Issue 12
December 1996
Pages 1533-1536

Full Paper *Chem. Eur. J.* **1996**, *2*, *No.12*, p.1533-1536

Ruthenium(II)-Catalyzed Oppenauer-Type Oxidation of Secondary Alcohols

Dr. Maria L. S. Almeida, Dr. Marion Beller, Dr. Guo-Z. Wang, Prof. Jan-E. Bäckvall

First published: December 1996 | https://doi.org/10.1002/chem.19960021210 |



dinuclear complex

Table 1. Ruthenium-catalyzed oxidation of secondary alcohols to ketones by acetone [a].

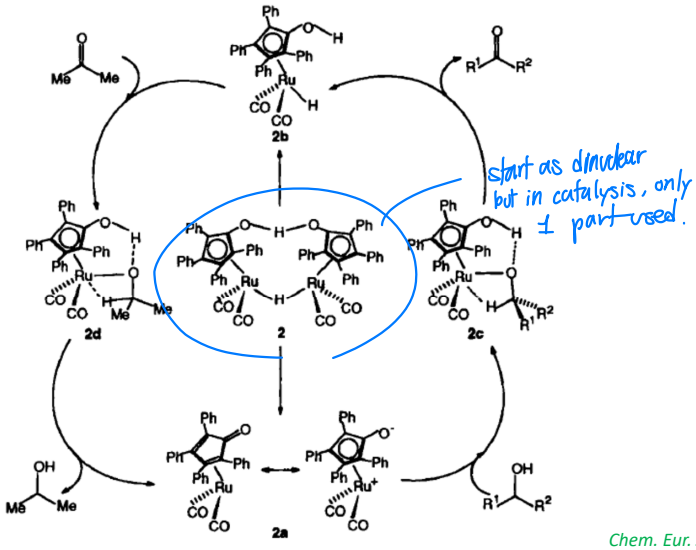
Entry	Substrate	Catalyst (mol %)	Time (h)	Product	% Yield [b]
1		1 (0.1)[c]	1.5		90
2		1 (0.2)	1		99[d]
3	3	2 (0.1)	17	14	87
4	 4	1(0.1)	7	 15	90
5		1 (0.1)[e]	3		87
6	5	2 (0.1)	22	16	92

much longer

[a] Unless otherwise stated, the reactions were carried out on a 1–5 mmol scale in refluxing acetone (1–5 mL) under N_2. With catalyst **1**, 0.1 equiv of K_2CO_3 is used. [b] Isolated yields after column chromatography unless otherwise noted. [c] Performed on a 50 mmol scale in 50 mL of acetone. [d] Determined by GC. [e] Performed on a 120 mmol scale in 100 mL of acetone.

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Dinuclear Ru catalyst in oxidation of alcohols



start as dinuclear but in catalysis, only 1 part used.

Scheme 3. Mechanism for the dehydrogenation of alcohols by acetone in the presence of complex **2**.

Chem. Eur. J. **1996**, *2*, *No.12*, p.1533-1536

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^{1}H & ^{31}P NMR in Inorganic Chemistry

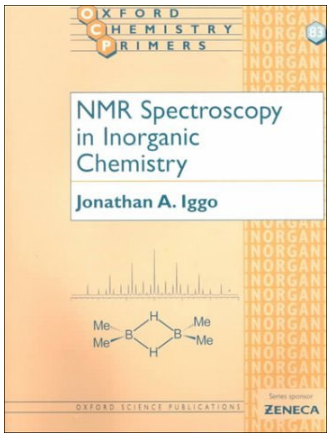
This topic is related to "Experiment 1: Homogenous Catalysis & Experiment 2: Metallocene".

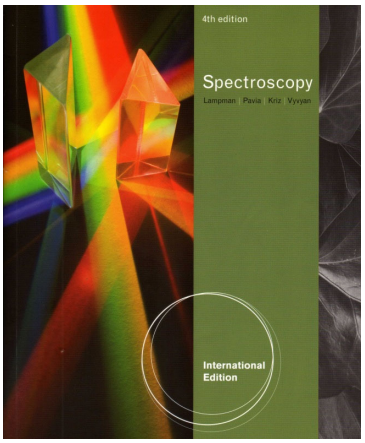
Disclaimer: All materials, information, weblinks and pictures contained in this notes are for educational purposes only.

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^{1}H-NMR in inorganic chemistry

References:





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^{1}H: Nuclear Spin I = ½

- These nuclei do interact with an applied magnetic field and are NMR active.
- The most commonly encountered nuclei: ^{1}H (ordinary hydrogen) and ^{13}C (1% abundance). Over 90% of all NMR spectra reported in the chemical literature are ^{1}H or ^{13}C NMR. →determining the structures of unknown organic and inorganic compounds.
- Other commonly observed nuclei: ^{19}F and ^{31}P. The analysis and interpretation of NMR spectra for I = ½ nuclei are based on a common set of rules, independent of the nature of the element.
- With I = ½, the spin-spin coupling behaviour is using the (n + 1) rule, which is originally from (2nI + 1) rule to calculate the number of lines observed when there is coupling to n equivalent nuclei.

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^{1}H-NMR in inorganic chemistry

- identify the structures of inorganic or organic compounds, with the use of the following features

1. number of signals
   → number of set of chemically equivalent protons
2. position of the signals
   → chemical environment of protons → chemical shift
3. intensity of signals
   → relative number of protons of each type
4. splitting of signals
   → connectivity of protons within a molecule

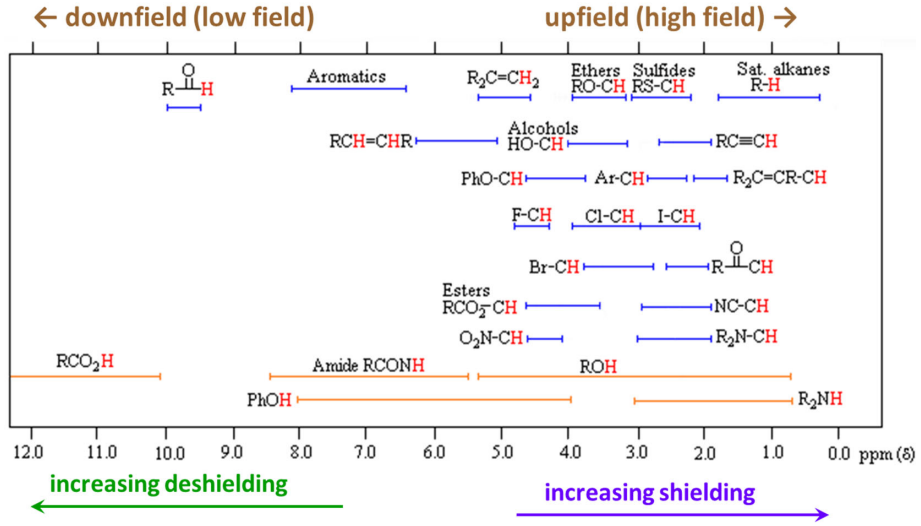
- Other factors to be considered for inorganic compounds: (i) oxidation state; (ii) nature of the ligands; (iii) nature of bonding; (iv) charge; (v) geometry (vi) nuclear spin and natural abundance of the neighbouring NMR active nuclei.

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^{1}H-NMR: chemical shift value

δ diff if diff ref & sol

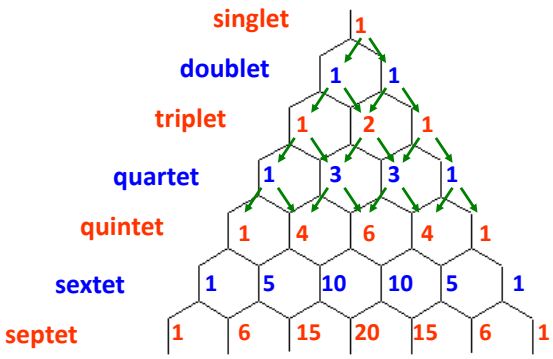
Remarks: δ scale is relative to TMS at δ = 0. Samples are characterized in $CDCl_3$ solution.



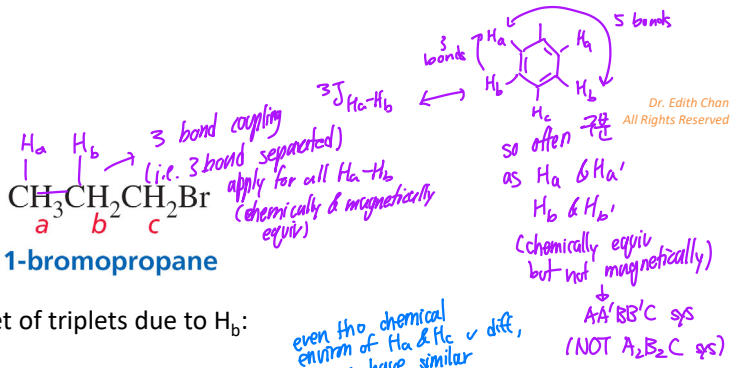
source of picture: http://www.cem.msu.edu

Splitting of signals

- **When *I* = ½, n + 1** rule follows the entries in **Pascal's triangle**:

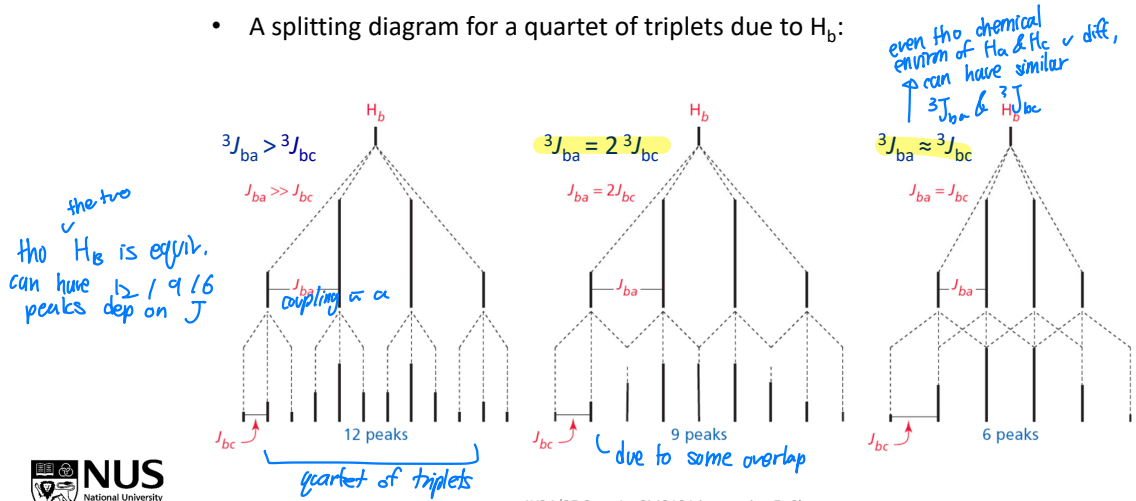


^{1}H-NMR: coupling constant



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- A splitting diagram for a quartet of triplets due to H_b:



First-Order Spectra

(predict # of splitting)

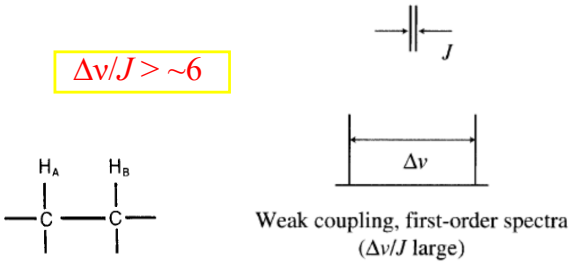
-Weakly coupled system;

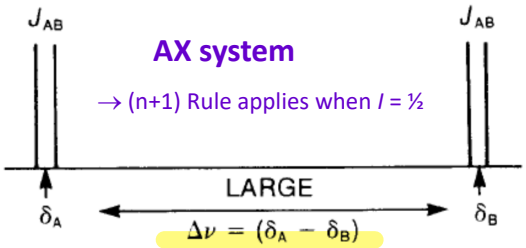
- Pople notation used: X, Y, Z
  → protons of widely different chemical shift values

- If 3 protons having very different chemical shifts, are involved, it is called AMX system;

- If 2 protons having same chemical shift are magnetically inequivalent, the system is designated as AA'.

$\Delta\nu/J > \sim 6$





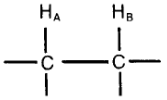
Second-order Spectra

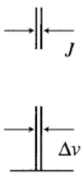
L CAN'T vs (n+1) rule to explain the spectrum

$\Delta\nu/J < \sim 6$

- strongly coupled system;
  ΔV is very small
- Pople notation used: A, B, C
  → protons of SIMILAR chemical shift values
- If 2 protons having SIMILAR chemical shifts, are involved, it exhibits second-order splitting and it is called AB system;
- (n+1) rule do not apply, and the difference in the chemical shift between H_A and H_B must be calculated:

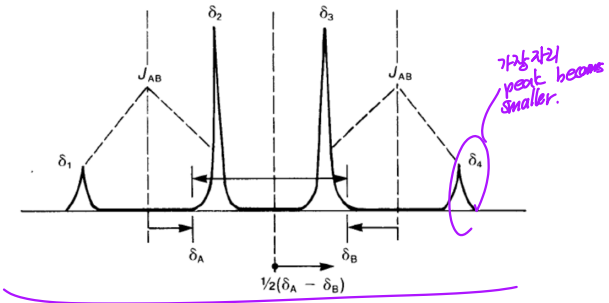
$$(\delta_A - \delta_B) = \sqrt{(\delta_1 - \delta_4)(\delta_2 - \delta_3)}$$





Strong coupling, second-order spectra ($\Delta\nu/J$ small)

AB system



가운데 2개 peak becomes smaller.

NOT quartet, just 2 doublets interacting

Reference: Donald L. Pavia, *Introduction to Spectroscopy*, Brooks/Cole, Cengage Learning, 2009.

Second-order Spectrum

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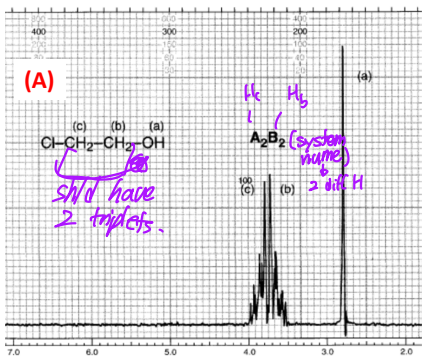
$$\delta\ (\text{ppm}) = \frac{\nu_{\text{sample}} - \nu_{\text{reference}}\ (\text{Hz})}{\nu_{\text{spectrometer}}\ (\text{Hz})} \times 10^6 = \frac{\Delta\nu\ (\text{separation from reference in Hz})}{\nu_{\text{spectrometer}}\ (\text{MHz})}$$

- **How to reduce Second-Order effect?**

Example: ^{1}H NMR Spectra of 2-chloroethanol.

Spectrum **A**: At 60 MHz.
Spectrum **B**: At 300 MHz.
Spectrum **C**: At 500 MHz.

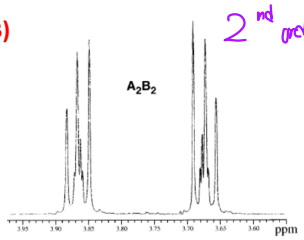
↑ operating freq
↓
↑ resol



Hc Hb
A_2B_2 (system name)
2 diff H

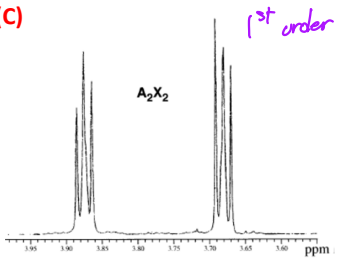
(c) and (b) should have 2 triplets.

(B)



2nd order

(C)

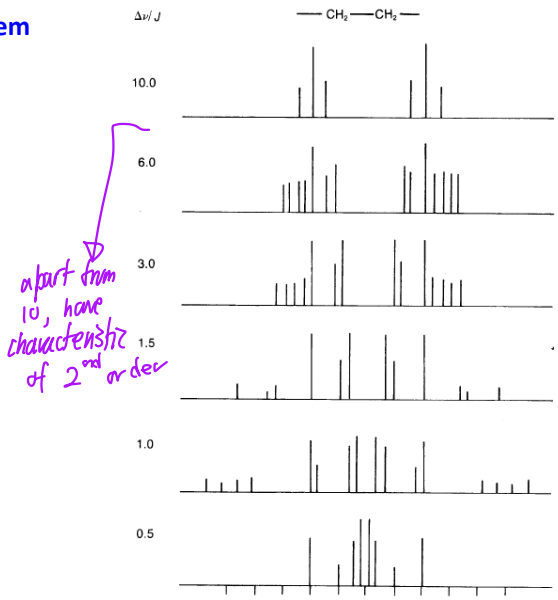


1st order

Reference: Donald L. Pavia, *Introduction to Spectroscopy*, Brooks/Cole, Cengage Learning, 2009.

Second-order Spectra

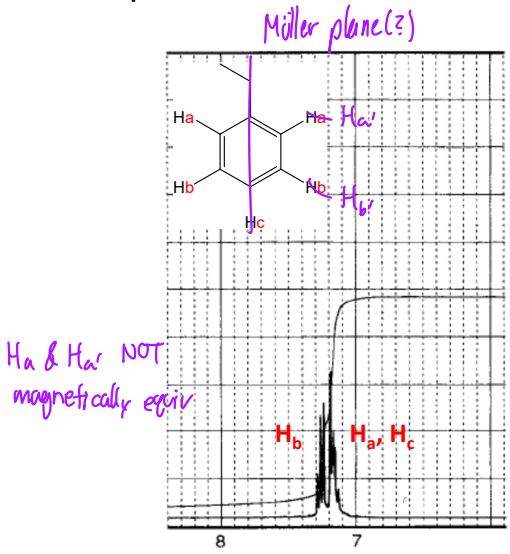
Four-proton system



Reference: Donald L. Pavia, *Introduction to Spectroscopy*, Brooks/Cole, Cengage Learning, 2009.

^{1}H NMR spectrum of aromatic compounds

Magnetic inequivalence



Reference: Donald L. Pavia, *Introduction to Spectroscopy*, Brooks/Cole, Cengage Learning, 2009.

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Scalar coupling in inorganic chemistry

- Scalar coupling between different NMR active nuclei present in inorganic molecule will occur. Homonuclear coupling is the coupling between the nuclei of the same element. Heteronuclear coupling is the coupling between the nuclei of the different elements.

- origin: the nuclear magnetic moments of other different NMR active nuclei surrounding the nucleus, which is under observation, produce "extra" magnetic fields;

- The magnitude depends on the interaction of the nuclear magnetic dipoles only;

- Decoupling = Removal of scalar coupling: irradiation of the coupling nucleus at its resonance frequency to induce rapid transitions between the possible orientations of the spin on the coupling nucleus;
  Example: $^{31}P\{^{1}H\}$ NMR spectrum = proton decoupled phosphorus-31 NMR spectrum

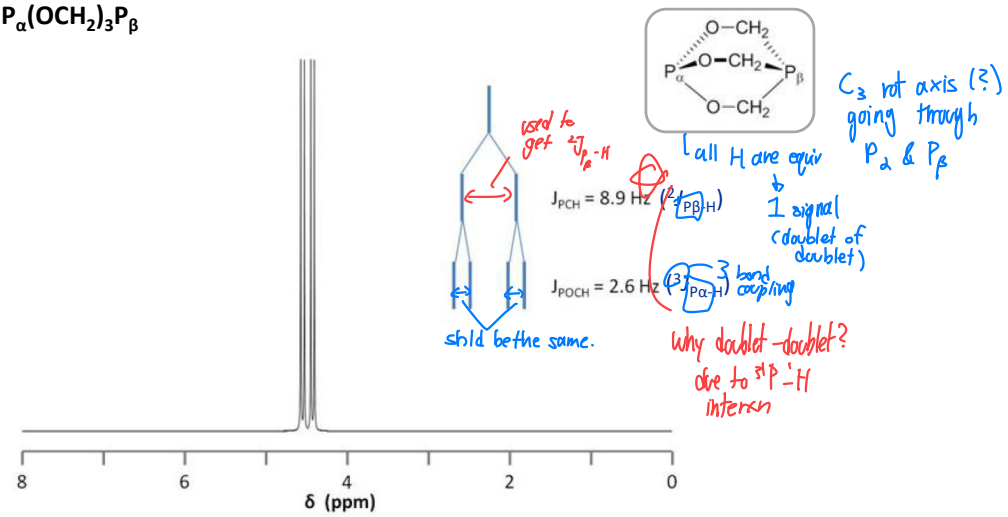
for simplified spec
can also use (n+1) rule
w/out coupling b/w $^{31}P-^{1}H$

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Scalar coupling: Examples

^{1}H NMR spectrum of $P_\alpha(OCH_2)_3P_\beta$

not ^{31}P



all H are equiv → 1 signal (doublet of doublet)

C_3 rot axis (?) going through P_α & P_β

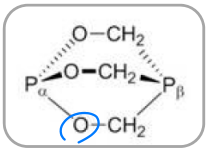
Why doublet-doublet? due to $^{31}P-^{1}H$ interaction

Source: http://cnx.org

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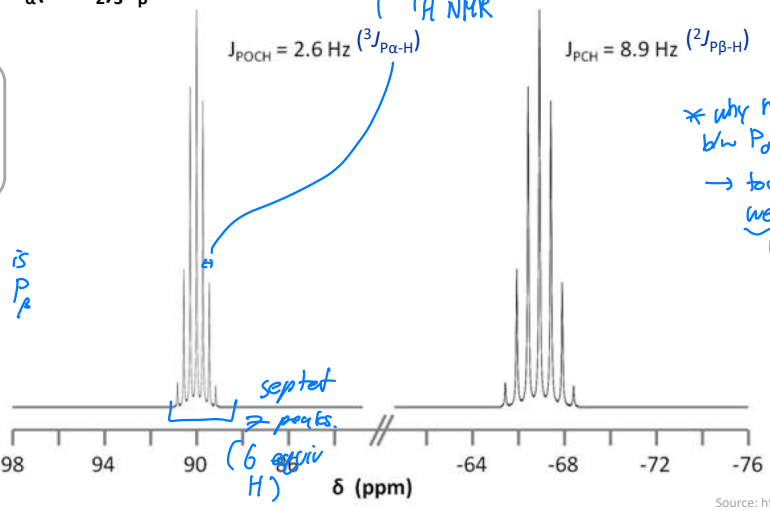
Scalar coupling: Examples

^{31}P NMR spectrum of $P_\alpha(OCH_2)_3P_\beta$



electro -ve. so S of P_α is higher than P_β

same as $^3J_{P_\alpha-H}$ found from ^{1}H NMR



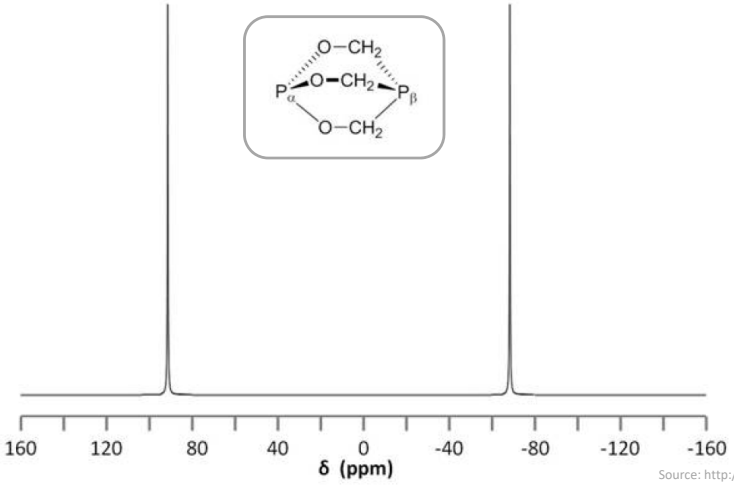
* why no coupling b/w P_α & P_β ?
→ too far. weak / no coupling
(may not be visible in spectrum

Source: http://cnx.org

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Scalar coupling: Examples

^{31}P{^{1}H} NMR spectrum of $P_\alpha(OCH_2)_3P_\beta$



Source: http://cnx.org

Remarks

- In general, the presence of Scalar couplings depends on:

(i) the relative abundance of the NMR active nuclei;

(ii) the absence of decoupling of the specific NMR active nuclei;

(iii) the strength of interaction between the two NMR active nuclei to be studied.